

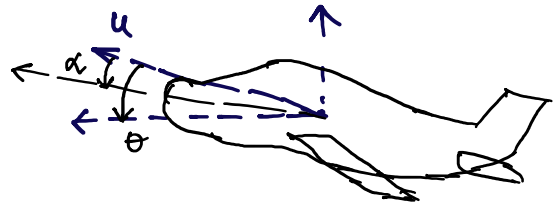
Example Description: A phantom Fighter

We're tasked with analyzing and stabilizing the longitudinal dynamics of a fighter jet at cruise. This is a fighter jet with a canard configuration referenced from Roskam[1] inherently with an unstable pole. We would like to assess this longitudinal model and design a controller to stabilize this aircraft, further we would like to ensure that this aircraft can track an enemy aircraft that's flying at a perturbed speed of 10ft/s and finally try to match the speed of an enemy aircraft at a perturbed speed of 100 ft/s in a specific timeframe with elevator constraints.

Mathematical Model :

The longitudinal motion involves perturbations in forward velocity (u), angle of attack (α), pitch rate (q), and pitch angle (θ), controlled by the elevator deflection δ_e . The book provides the dynamics model of the aircraft as given below.

$$\begin{aligned}\dot{u} &= -g\theta \cos \theta_1 + (X_u + X_{T_u})u + X_\alpha \alpha + X_{\delta_e} \delta_e \\ U_1 \dot{\alpha} - U_1 q &= -g\theta \sin \theta_1 + Z_u u + Z_\alpha \alpha + Z_{\dot{\alpha}} \dot{\alpha} + Z_q q + Z_{\delta_e} \delta_e \\ \dot{q} &= (M_u + M_{T_u})u + (M_\alpha + M_{T_\alpha})\alpha + M_{\dot{\alpha}} \dot{\alpha} + M_q q + M_{\delta_e} \delta_e \\ \dot{\theta} &= q\end{aligned}$$



This is further converted into state-space with the resulting equation (the dynamic derivative constants are provided on table 5.5 of Roskam[1])

$$\begin{aligned}x &= [u \quad \alpha \quad q \quad \theta]^T & \dot{x} &= Ax + Bu \\ \dot{x} &= \begin{bmatrix} -0.0019 & -6.2907 & 0 & -32.2 \\ -0.0001175 & -0.4336 & 0.9972 & 0 \\ -0.0007968 & 10.5511 & -0.2629 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} x + \begin{bmatrix} -0.2834 \\ -0.0407 \\ 6.6717 \\ 0 \end{bmatrix} \delta_e\end{aligned}$$

Controllability test shows that this system is controllable (LQR.mlx) and eigenvalue calculations show the following 4 poles with one unstable pole at 2.8993

$$[2.8993, -3.5925, -0.0026 + 0.0700i, -0.0026 - 0.0700i]$$

As a result, we will have to design a controller such that it will trim the aircraft in cruise and stabilize it. We will leverage a LQR (Linear Quadratic Regulator) to ensure that this aircraft is trimmed in cruise, further we will ensure that the Aircraft can track another aircraft at a constant speed with a Linear Quadratic Tracking controller. This Aircraft is modeled as a LTI (Linear Time Invariant) problem where the system dynamics remains same over time, A and B are constant.

Linear Quadratic Regulator[2] :

Cost functions:

$$\begin{aligned}J &= 1/2 \int_0^\infty (x^T(t)Qx(t) + u^T(t)Ru(t))dt \\ Q &= I4 \text{ (Identity Matrix)}, R = 0.1\end{aligned}$$

Leveraging the LQR function in Matlab, we find the gains and then subsequently the closed loop eigenvalues as follows. The LQR function solves the Algebraic Riccati Equations (ARE) to calculate the gain.

Closed loop poles : $[-10.9280; -3.7627+4.3212i; -3.7627-4.3212i; -0.3127]$, all the poles are stable as they are on the left half planes.

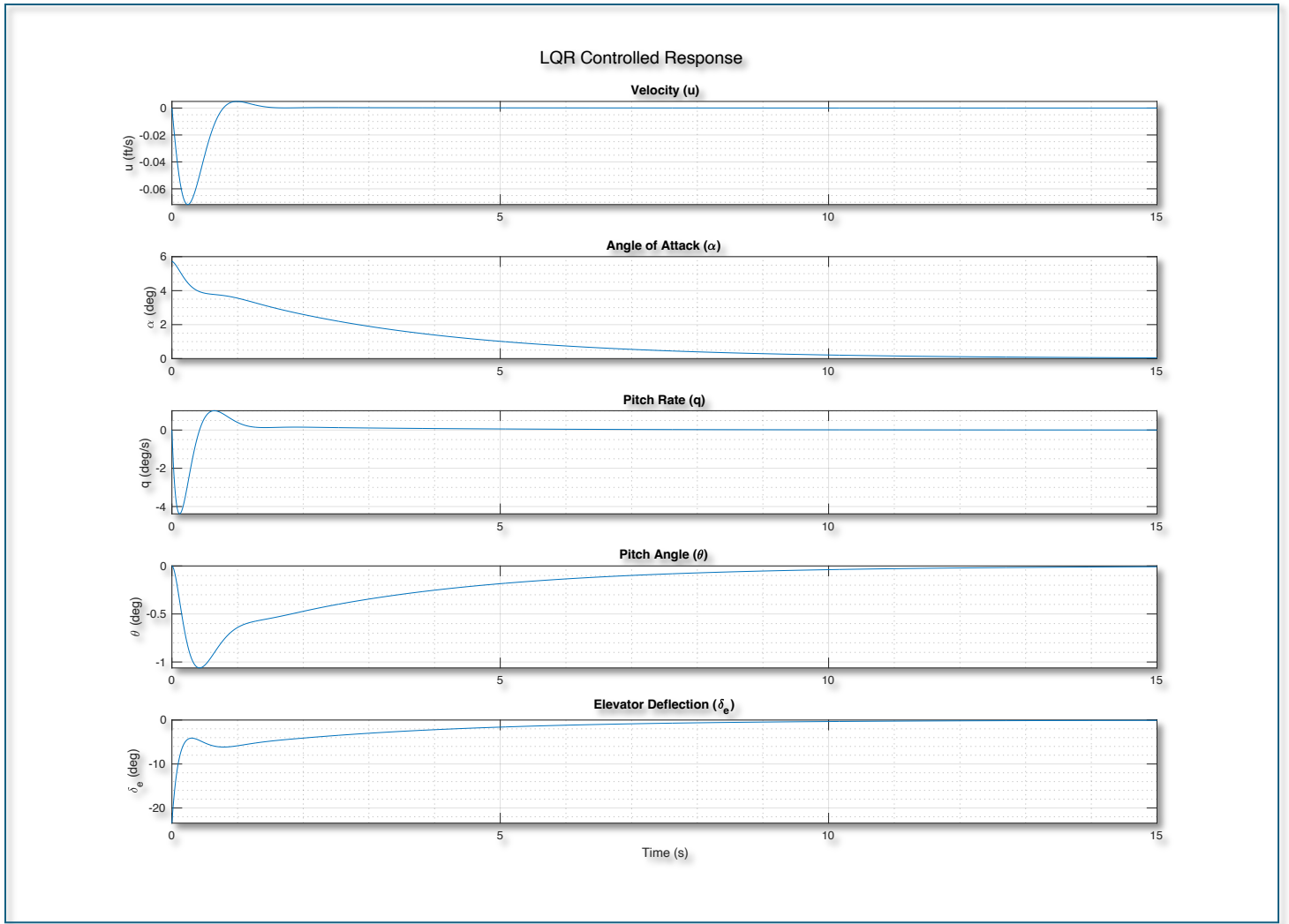


Figure 1 : LQR simulation results

From the plots above we can see that this controller is able to trim the aircraft , this is a response to 0.1 radians (5.7 degrees) angle of attack initial condition.

Assumptions : We assume that the dynamics model is perfect, and that full state feedback is available, unlimited control , no saturation of actuators is also assumed. Finally, this optimization is performed over an infinite time horizon.

Linear Quadratic Tracker (LQT)[2]

Now that we can trim this aircraft in cruise and fly , we would like to see if we can track an enemy aircraft that's flying at constant but perturbed speed of 10ft/s compared to our jet.

A and B matrix remain same above , we will use $C = [1 \ 0 \ 0 \ 0]$ since we want to track $u(t)$ which is the first state. We will define an internal model to account for the reference velocity , $z(t) = 10\text{ft/s}$, $Z(s) = 10/s$ (Laplace Transform)

$$\dot{x}(t) = Ax(t) + Bu(t), \quad y = Cx(t)$$

Let's introduce $z(t) = 10 \text{ ft/s}$ as the reference velocity that needs to be tracked.

$e = z - y$ which gives $y = z(t)$, because we would like the error(e) to goto zero

Internal Model then becomes the following :

$$\dot{x}_m(t) = A_m x_m(t) + B_m e(t)$$

In CCF (controllable Canonical Form) , a step reference becomes

$$A_m = [0], B_m = [1]$$

The Augmented system now becomes the following :

$$\tilde{x} = \begin{bmatrix} x \\ x_m \end{bmatrix} = \begin{bmatrix} A & 0 \\ -B_m C & A_m \end{bmatrix} \tilde{x} + \begin{bmatrix} B \\ 0 \end{bmatrix} u + \begin{bmatrix} 0 \\ B_m \end{bmatrix} z$$

Let's use a quadratic objective function of the following type

$$J = 1/2 \int_0^\infty (\tilde{x}^T(t) Q \tilde{x}(t) + u^T(t) R u(t)) dt$$

here $Q = \text{diag} [1,1,1,1,25]$, 25 is the weight on the tracking error , $R = 100$, had to increase R such that the elevator deflections were reasonable (< 40 degrees) , trial and error method.

Now we design the controller in Matlab (LQT.mlx) using the LQR function leveraging the newly Augmented system and then we simulate, the results are shown below, the aircraft can track the reference velocity and stabilizes within 10 seconds.

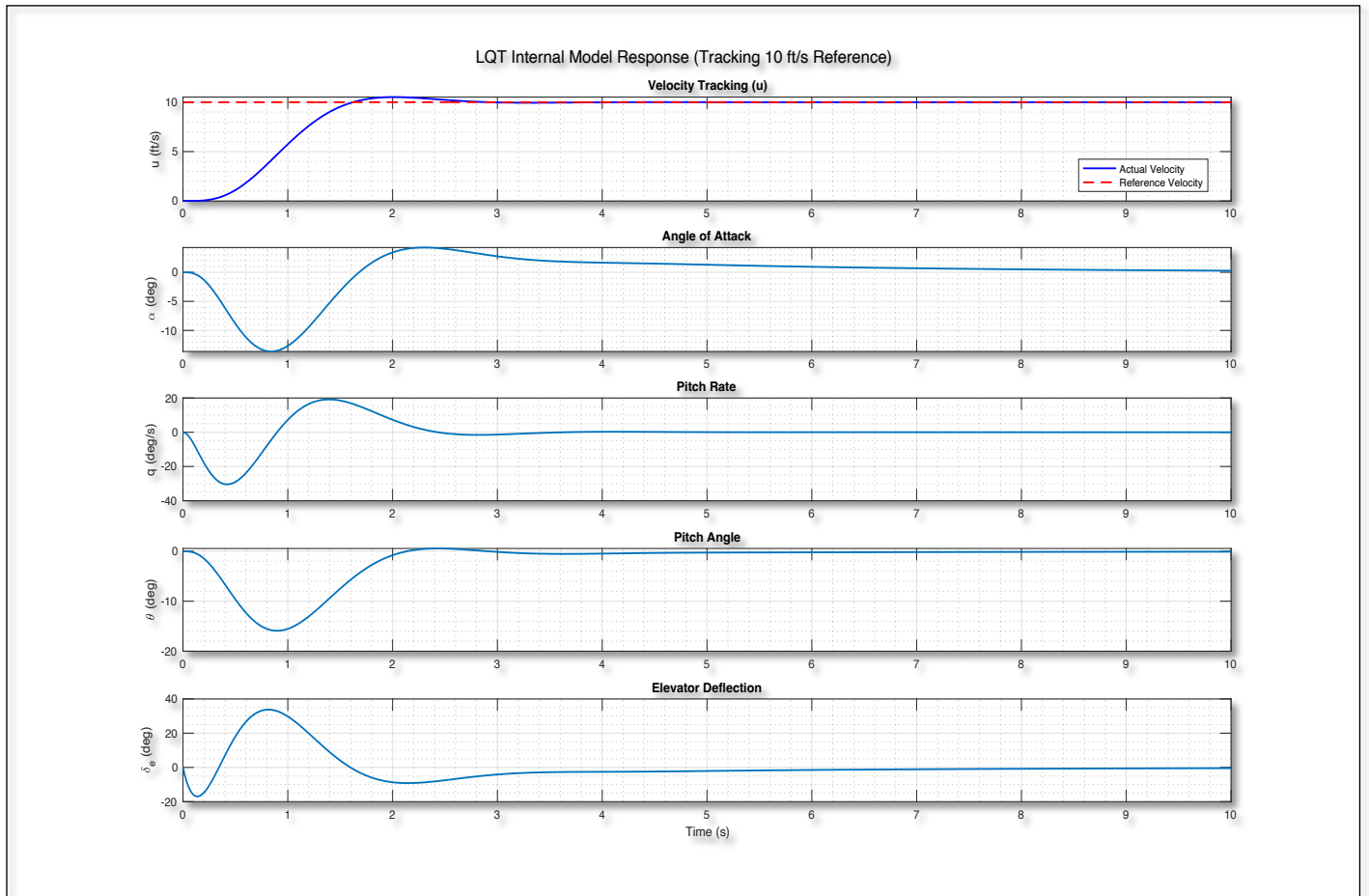


Figure 2 : LQT simulations results

Assumption made on the LQT system :

Model uncertainty is ignored, linear LTI system , full state is available, a constant reference signal is assumed (not always practical) and finally no input or thrust constraints are incorporated.

Comparison between LQT and LQR method and recommendation.

LQR effectively stabilizes aircraft and regulates states to zero, maintaining trimmed flight conditions over an infinite horizon but lacks the ability to track non-zero commands. LQT extends LQR by incorporating reference inputs, enabling precise tracking of constant references, such as an enemy aircraft's steady 10 ft/s velocity, by minimizing tracking error. For tasks requiring constant-speed target tracking, *LQT is the superior and recommended method due to its specific design*

PMP (Pontryagin's Minimum Problem) [2]:

We have designed a regulator to keep the aircraft trimmed during cruise (LQR) , we have also designed the tracking system to ensure that the aircraft is able to track an enemy aircraft at a constant speed, next we would like to design a controller that can intercept an enemy aircraft in a given amount of time while keeping in mind the some physical control constraints (Elevator Deflections of ± 35 degrees) .

$$\dot{x}(t) = Ax(t) + Bu(t) \text{ } A \text{ and } B \text{ are the same as the original problem}$$

$$\text{Initial state: } x(0)=[0,0,0,0] \text{ Final state: } x(8)=[100,0,0,0]$$

$$\text{Control Constraint: } u \in [-0.6109, 0.6109] \text{ (35 degrees in radians)}$$

$$J = \frac{1}{2} \int_0^5 \left(x^T(t) Q x(t) + u^T(t) R u(t) \right) dt, \text{ } Q = \text{diag}(1, 75, 100, 1) \text{ and } R = 0.5$$

I came up with this objective function by trial-and-error method. Had to penalize the angle of attack and pitch rate to keep them relatively lower.

$$\text{Hamiltonian : } H = \frac{1}{2} (x^T(t) Q x(t) + R u(t)^2) + \lambda^T (A x(t) + B u(t))$$

$$\frac{\delta H}{\delta u} = 2 * R * u + B * \lambda^T$$

$$u^* = -(B/2R) * \lambda^T \text{ (used in the code)}$$

$$\dot{\lambda}(t) = - \frac{\delta H}{\delta x}$$

$$\dot{\lambda}(t) = -Qx(t) - A^T \lambda(t) \text{ (Used in the code)}$$

$$u = \max(-0.6109, \min(0.6109, -(1/R)B^T \lambda))$$

This model of the aircraft doesn't have the thrust incorporated, so we are assuming that the aircraft is providing constant thrust to execute this maneuver , we see that the aircraft must make some pretty drastic maneuvers to come up with the control logic. A better approach would be to incorporate a thrust control to avoid such drastic maneuvers.

I leveraged a numerical solution to get better results for this aircraft as this is a 4th order system . This is a 4th order system and as we know we can only have 3 switching ; this was verified by the results below.

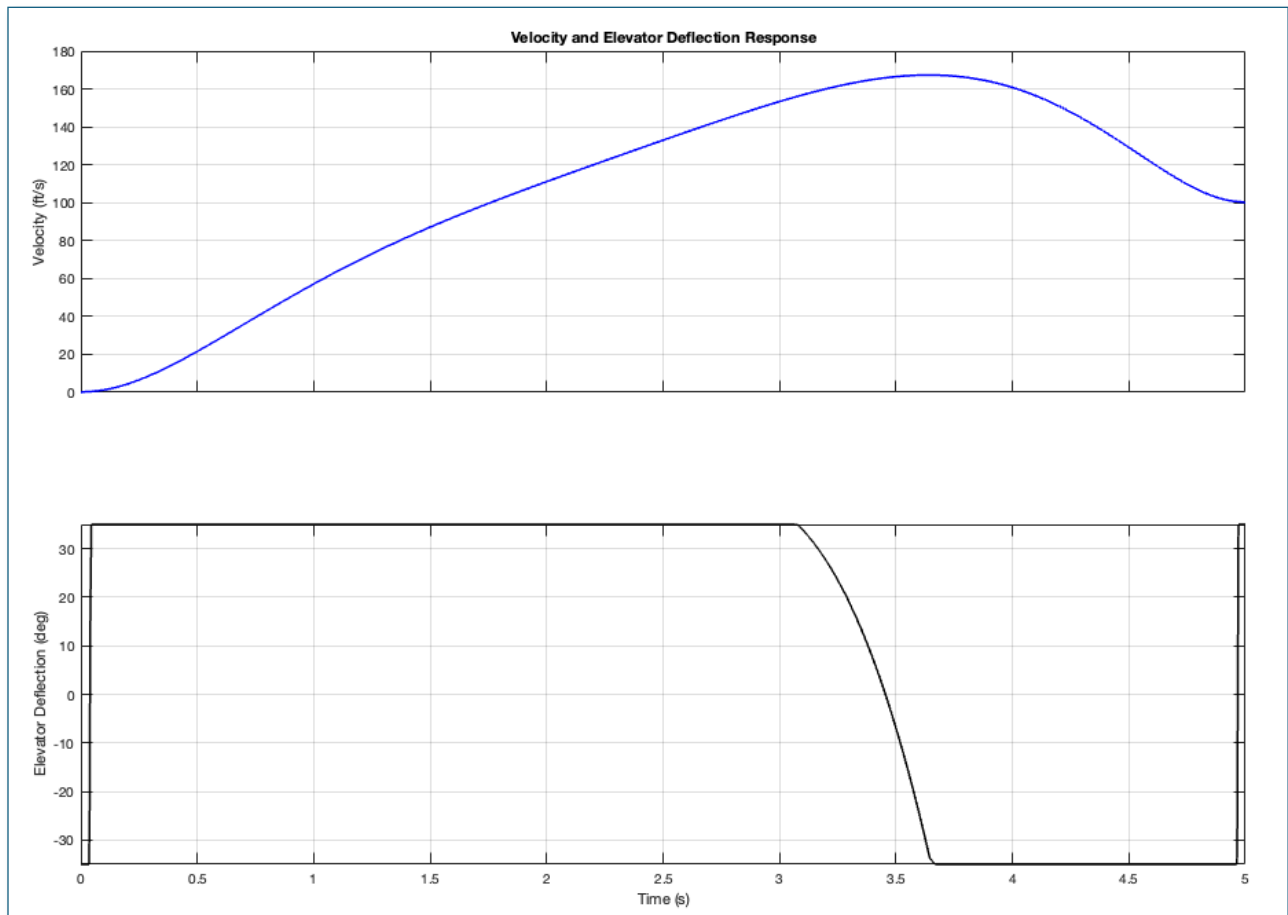


Figure 3 : PMP simulation results

Comparison between LQT,LQR and PMP method and recommendation.

The investigation of three control design methodologies for a fighter jet's longitudinal dynamic reveals that LQR stabilizes the aircraft around a trim condition over an infinite horizon, while LQT extends LQR to track constant non-zero velocities, such as an enemy aircraft's steady speed. However, neither LQR nor LQT is designed for precise state targets at a fixed time or limited control authority. For intercepting an enemy aircraft by achieving a specific velocity (100 ft/s) within 5 seconds while respecting elevator deflection limits, PMP works. *PMP is the most suitable and recommended approach for aggressive maneuvers under strict physical and time constraints for a fighter jet.*

Method	LQR	LQT	PMP
Purpose	Regulate to Zero/Trim	Track Constant Reference	Finite-Time Optimal Maneuver
Time Horizon	Infinite	Infinite	Finite (5)
Terminal State	Free (converges to 0)	Free (tracks reference)	Fixed (100, 0, 0, 0) at t_f
Control Constraints	none	none	Yes
Report Application	Stabilize cruise trim	Track 10 ft/s constant vel.	Intercept 100 ft/s with $\pm 35^\circ$ elev.

Reference:

- [1] J. Roskam, *Airplane flight dynamics and automatic flight controls*. Lawrence, Kan: DARcorporation, 2003.
- [2] D. S. Naidu, *Optimal Control Systems*, 1st edition. Boca Raton, Fla.: CRC Press, 2002.